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Exploring two-trace two-dimensional (2T2D) correlation spectroscopy as an effective approach to improve accuracy of discriminant analysis by highlighting asynchronous features in two separate spectra of a sample

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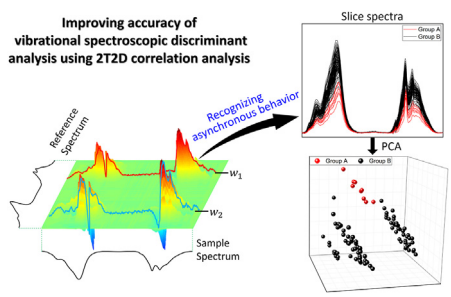
HIGHLIGHTS

- A rapid bile-based IR spectroscopic screening of gall bladder (GB) cancer was demonstrated.
- 2T2D correlation analysis highlights asynchronous features in two separate spectra of a sample.
- The asynchronous features would be thereby more sample-to-sample characteristic and discriminant.
- The use of 2T2D slice spectra improved the accuracy of olive oil authentication up to 90.0%.
- The accuracies for identifying the geographical origins of agricultural products were also elevated.

GRAPHICAL ABSTRACT



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ABSTRACT

This study aims to demonstrate two-trace two-dimensional (2T2D) correlation spectroscopy as an effective tool for improving the accuracy of discriminant analysis. Because 2T2D correlation analysis allows sensitive capturing of asynchronous spectral behaviors between two compared spectra of a sample, the subsequent asynchronous correlation features are expected to reveal more sample-to-sample characteristics and discriminants than the original spectral feature. Initially, near-infrared (NIR) spectroscopic authentication of pure olive oil was performed using the spectra collected at 20 °C and 41 °C. When the 2T2D slice spectra of the samples were used for the discriminant analysis, the authentication accuracy reached to 100%, while became degraded in the cases of using the spectra collected either at 20 °C or 41 °C. Furthermore, a simple strategy of utilizing the average spectrum of one sample group as the reference spectrum in the 2T2D correlation analysis was proposed for two-group discrimination and evaluated for the NIR identification of the geographical origins of agricultural products (milk vetch root (MVR) and perilla seed samples). Because the average spectrum of one sample group was used for comparison, dissimilar constituent compositions of the samples in another group were better observed, thereby improving the accuracy of discrimination of the geographical origins of the samples in both cases. The overall results demonstrated that 2T2D correlation analysis is effective for

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highlighting the minute asynchronous spectral features of a sample and can be expanded for diverse vibrational spectroscopy-based discriminant analyses.

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1. Introduction

Two-trace two-dimensional (2T2D) correlation spectroscopy [1] utilizing only a pair of spectra instead of a series of spectra, which are collected under external perturbation, was recently proposed by Noda as a simple and alternative approach to 2D correlation analysis. 2T2D correlation analysis synchronizes the intensity changes of the spectral bands that are unable to vary independently of each other in two compared spectra, consequently rendering the concurrently occurring spectral variations silent. Therefore, potential asynchronous spectral behaviors arising from different sources in a pair of compared spectra can be more sensitively highlighted. Moreover, the broadly overlapping and barely discernible neighboring bands are better featured as long as their origins are dissimilar. Owing to this advantage, 2T2D correlation analysis has been recently adopted as a tool for the spectroscopic investigation of the structural changes in several samples. Some examples are the near-infrared (NIR) rheo-optical characterization of hydroxyl-functionalized polypropylene silica nanocomposites [2], UV–Vis investigation of the coordination between organic ligands and Ni^{2+} ions [3], and infrared (IR) investigation of adulterants in dietary supplements [4].

Instead of being used as a supporting qualitative spectroscopic analysis tool, as was typical in previous research, in this study, 2T2D correlation analysis is explored as a direct and effective sample-to-sample discriminator, capable of improving the accuracy of discriminant analysis. When the sample spectral features are highly overlapping and indistinct owing to complex chemical compositions, sample distinction becomes highly difficult, and the accuracy is thus expected to degrade. In this case, a method that can identify minute dissimilarities in the spectral features of the samples is critical to maximize the accuracy of discriminant analysis. 2T2D correlation analysis is a promising option to meet the above demand because it compares a sample spectrum with a reference spectrum for capturing the asynchronous spectral behaviors in both spectra. Therefore, the acquired asynchronous correlation features of a sample are expected to be more characteristic of the individual samples than simply a raw spectral feature. This potential has motivated us to newly incorporate 2T2D correlation spectroscopy in discriminant analysis for enhancing its accuracy.

For the first study, previous research on the NIR spectroscopic authentication of pure olive oils utilizing the power spectra obtained from the 2D correlation analysis of temperature-varied (20, 27, 34, and 41 °C) spectra [5] was recalled. The features of the power spectra were more sample-distinguishable and aided in improving the authentication accuracy compared to that from the constant-temperature measurement. The temperature-induced spectral variation is expected to be more sensitively identified by 2T2D correlation analysis, because it is based on the direct comparison of two selected spectra, instead of trend-tracking in a series of temperature-varied spectra in conventional 2D correlation analysis. Therefore, 2T2D correlation analysis was performed using sample spectra collected at 20 °C and 41 °C. Following this, the corresponding 2T2D asynchronous correlation slice spectra were used for authentication. *k*-Nearest neighbor (*k*-NN), partial least squares-discriminant analysis (PLS-DA), and support vector machine (SVM) were employed for the discrimination and the subsequent accuracies were assessed in comparison with those in the measurements at 20 °C and 41 °C.

In a meanwhile, in the vibrational spectroscopic discriminant analysis of solid samples [6–9], e.g., using diffuse reflectance measurements, temperature perturbation (control) of a sample is difficult and also unrealistic in practical routine analysis. To implement 2T2D correlation analysis in this case, a simple and alternative strategy of utilizing the average sample spectrum in one group as the reference spectrum in 2T2D correlation analysis was proposed for two-group discrimination. Because the average spectrum of one sample group is used for comparison, dissimilar constituent compositions of samples in another group are expected to be more sensitively highlighted with 2T2D correlation analysis. For the evaluations, NIR spectroscopic discriminations of the geographical origins of two different agricultural products, milk vetch root (MVR) and perilla seed samples, were conducted. 2T2D correlation analysis was performed using each sample spectrum with respect to the determined average spectrum, and the obtained 2T2D slice spectra were used in the identification of the geographical origins of the samples. The resulting discrimination accuracy was assessed and compared to those for using raw spectra.

2. Experimental

2.1. NIR spectral dataset of pure and adulterated olive oil samples

The NIR spectra of 9 pure and 90 adulterated olive oils, including blends with canola, soybean, and corn oils (adulteration rate: 5.0%), were employed as in a former research [5]. The spectra of all the samples were collected at 20 and 41 °C. *k*-NN, PLS-DA, SVM were adopted for the authentication. Two-level cross validation (CV) composed of inner and outer CV loops [10,11] was used in this study to evaluate authentication performance with 3-fold CV. The pure and adulterated samples were equally partitioned into 3 equal folds, respectively. In the inner CV loop, initially a calibration dataset was further divided into training and validation sets and an optimal classification model was searched by varying related parameters. In the outer CV loop, accuracy was obtained with an optimized model using test set. This whole procedure was iterated 100 times with random splits of the spectral dataset and then subsequent mean accuracy, mean sensitivity, and mean specificity were finally acquired. The parameters to determine for *k*-NN and PLS-DA were number of neighbors and number of principal component (PC) scores, respectively, and kernel function, cost (C), and gamma (γ) for SVM (refer to the supplementary information). All the discriminant analysis, principal component analysis (PCA), and 2T2D correlation analysis were executed using MATLAB R2011a (The MathWorks Inc., USA).

2.2. NIR spectral datasets of MVR and perilla seed samples

The NIR spectra of the MVR samples (25 domestic and 25 imported samples) acquired in a previous study [12] were adopted again. The NIR spectra of the perilla seed samples (a total of 245 samples, 154 domestic and 91 imported samples) were supplied by the National Agricultural Products Quality Management Service (NAQS, Korea), a certified government agent for the authentication

of the geographical origins of agricultural products. The samples were collected by NAQS over approximately 2–3 years to present diverse cultivation regions for the incorporation of wider compositional variations in the dataset. Each sample was ground into a powder (20-mesh), and the diffuse reflectance spectra of the powder samples were collected using a Fourier transform (FT)-NIR spectrometer (MATRIX-I, Bruker). All the NIR sample spectra were treated with multiplicative scatter correction (MSC) to reduce the unwanted spectral variations caused by scattering. Discrimination accuracies were obtained as the same fashion performed in the olive oil authentication, while, 5-fold partitioning of the dataset for CV was executed in the discriminant analysis of MVR and perilla seed samples.

3. Results and discussion

3.1. NIR spectroscopic authentication of pure olive oil using 2T2D correlation analysis

In a previous study, 2D correlation analysis was performed using the spectra collected at four different temperatures (20, 27, 34, and 41 °C), and the obtained power spectra were used for PCA. Fig. 1 (a) and (b) display the power spectra of all the pure and adulterated olive oil samples in the 8800–6600 cm^{-1} range as well as the subsequent three-dimensional (3D) scatter plot constructed with the first, second, and third PC scores, respectively. The red and black colors denote pure and adulterated samples, respectively. As depicted, the intensities in the adulterated sample power spectra are generally higher than those in the pure sample power spectra; however, a few of the adulterated sample power spectra are similar in shape to the pure sample power spectra. Therefore, a clear differentiation of the pure and adulterated samples is barely realizable. In the 3D score domain, the adulterated samples were generally divided into two groups and the pure samples were located between these two groups. A few of the pure samples were very close to the adulterated sample group at the left bottom.

Next, two spectra of each sample collected at 20 and 41 °C were used to perform the 2T2D correlation analysis. The spectra collected at these temperatures are referred as sample $s(\nu)$ spectrum and reference spectrum $r(\nu)$, respectively, and an asynchronous spectrum, $\Psi(\nu_1,\nu_2)$, which is a quantitative comparison of the spectral intensity variations observed at two different spectral variables, was obtained using equation (1).

$$\Psi(\nu_1,\nu_2) = \frac{1}{2}[s(\nu_1)\cdot r(\nu_2) - r(\nu_1)\cdot s(\nu_2)] \quad (1)$$

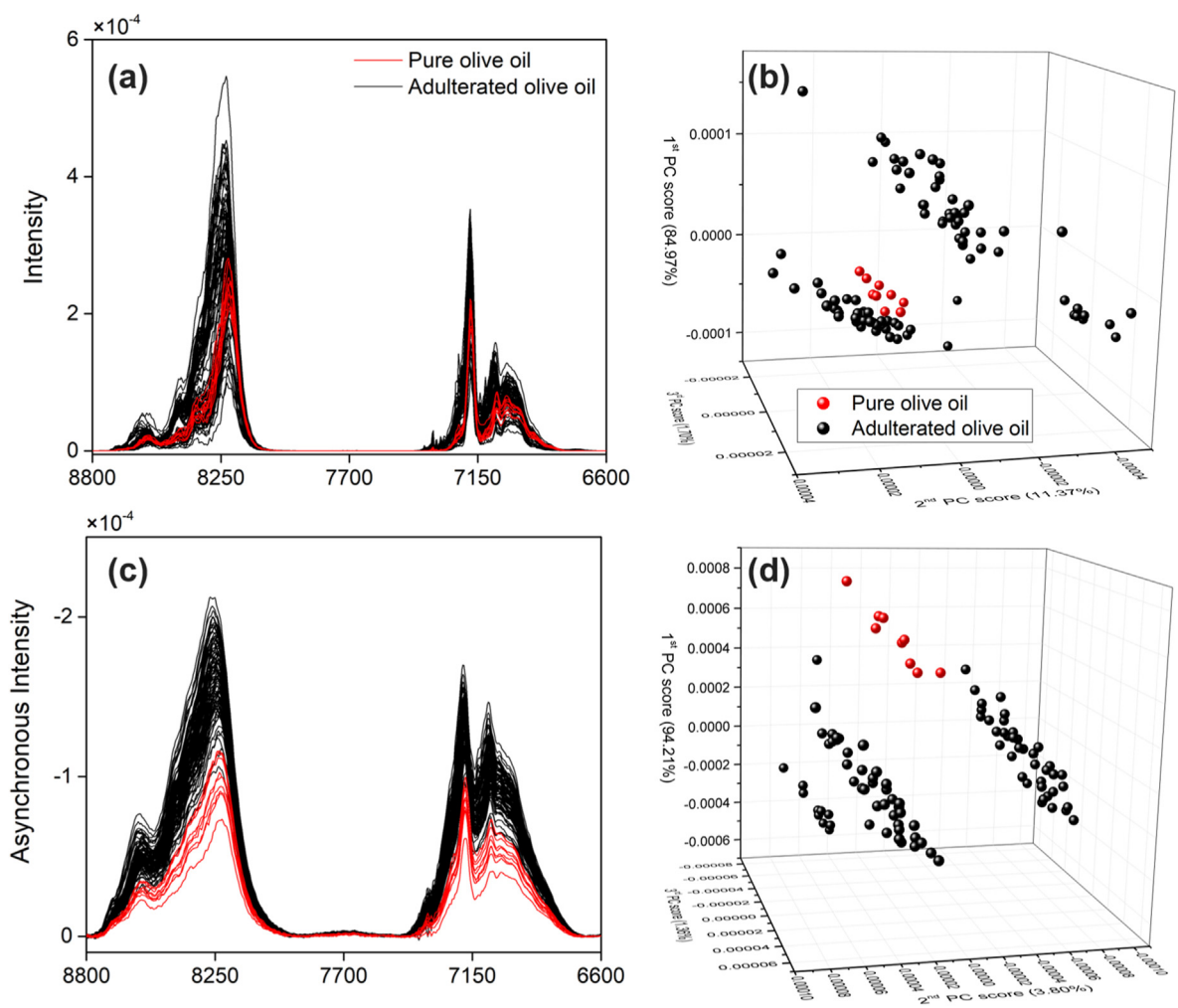

Fig. 1. Power spectra of all the pure and adulterated olive oil samples (a), and the subsequent 3D scatter plot constructed with the first, second, and third PC scores (b). In addition, the 2T2D slice spectra at 6665 cm^{-1} of all the olive oil samples (c) and the subsequent 3D score scatter plot (d) are shown. The red and black colors denote the pure and adulterated samples, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Fig. 2 presents the asynchronous 2T2D correlation plots obtained from olive oil #1 (a) and three adulterated samples commonly based on olive oil #1 (olive oil #1−canola oil #1 (b), olive oil #1−soybean oil #1 (c), and olive oil #1−corn oil #1 (d)). It can be easily noted that the asynchronous correlation pattern of pure olive oil #1 is remarkably different from those of the three adulterated samples. Asynchronous correlation peaks appear at approximately 8131 and 6765 cm^{-1} in the cases of the three adulterated samples, whereas they are absent in the case of olive oil #1. Concurrently, the intensities of the asynchronous correlation peaks differ among the adulterated samples.

Fig. 3 presents the asynchronous 2T2D correlation plots obtained from olive oil #2 and three adulterated samples commonly based on olive oil #2 (olive oil #2−canola oil #1 (b), olive oil #2−soybean oil #1 (c), and olive oil #2−corn oil #1 (d)). It can be seen that the asynchronous correlation pattern of olive oil #2 is nearly similar to that of olive oil #1 (Fig. 2 (a)), indicating the analogous temperature-dependent behaviors of the components of the pure olive oils. Concurrently, asynchronous peaks at approximately 8131 and 6765 cm^{-1} are only observed for the olive oil #2−soybean oil #1 (Fig. 3 (c)) and olive oil #2−corn oil #1 samples (Fig. 3 (d)), and not for the olive oil #2−canola oil #1 sample (Fig. 3 (b)). Overall, based on Figs. 2 and 3, 2T2D correlation analysis sensitively recognizes the adulteration and the asynchronous correlation patterns of the adulterated samples, which are different depending on the pair of the employed pure olive oil and adulterating oil.

Each 2T2D asynchronous correlation plot shown above comprises a total of 737 slice spectra, corresponding to 737 variables in the 8800−6600 cm^{-1} range. For detailed examination of 2T2D spectral feature sliced at a given wavenumber, the slice spectra at 6665 cm^{-1} of all the pure and adulterated samples were chosen as an example (Fig. 1 (c)) and the corresponding 3D score scatter plot (Fig. 1 (d)) was examined. Notably, in the slice spectra, the asynchronous intensities of the adulterated sample spectra (black) are higher than those of the pure sample spectra (red). This indicates that the degrees of dissimilar spectral behaviors in the two temperature-varied spectra of the adulterated samples are increased owing to the inclusion of the adulterant oils, which have temperature-dependent characteristics different from those of the pure olive oils. In the 3D score scatter plot, the pure samples are better isolated without overlapping with the adulterated samples of three separate groups. The pure and adulterated samples are more distinguishable using the slice spectra at 6665 cm^{-1},

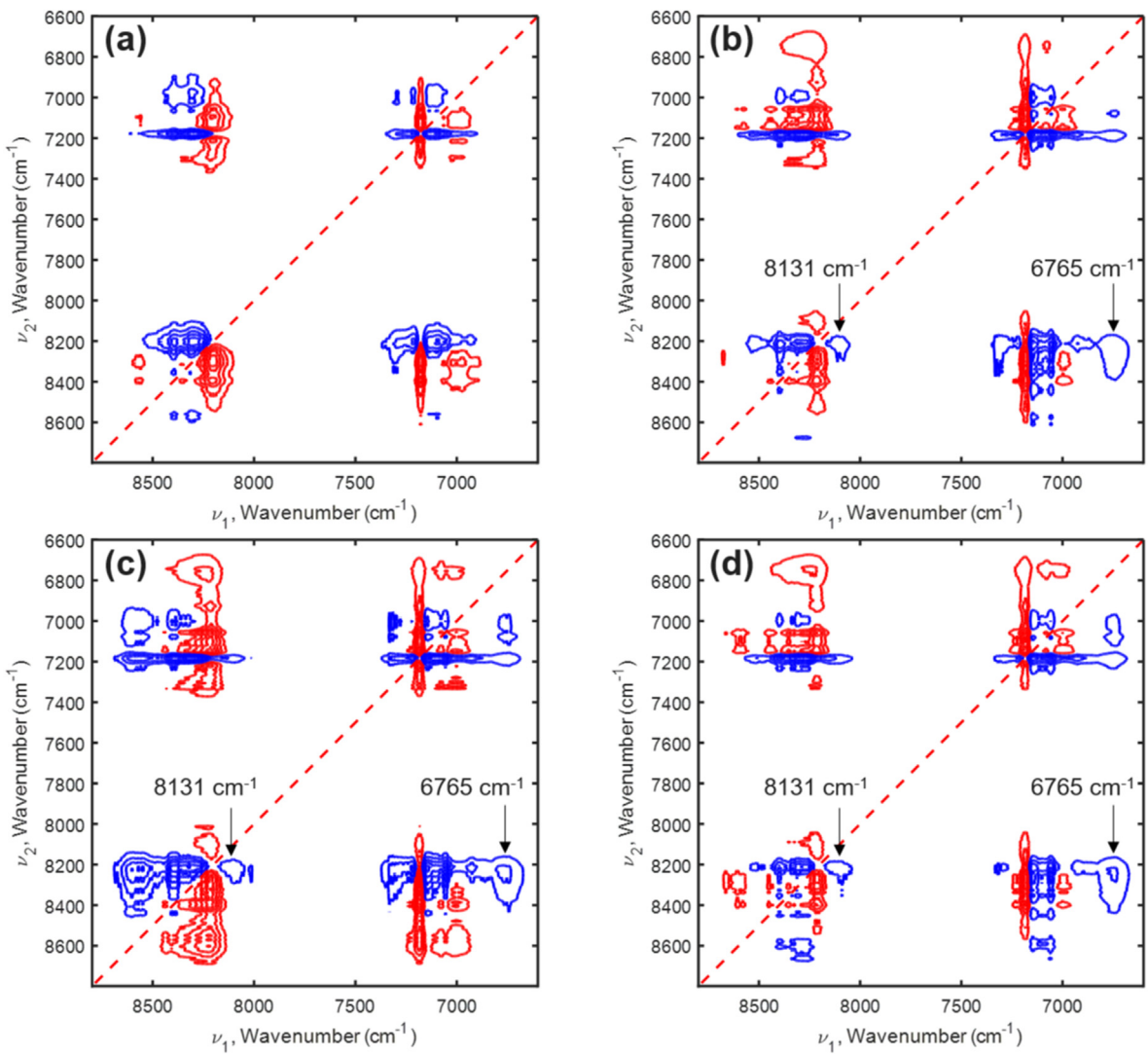

Fig. 2. Asynchronous 2T2D correlation plots obtained from pure olive oil #1 (a) and three adulterated samples commonly based on olive oil #1 (olive oil #1−canola oil #1 (b), olive oil #1−soybean oil #1 (c), and olive oil #1−corn oil #1 (d)).

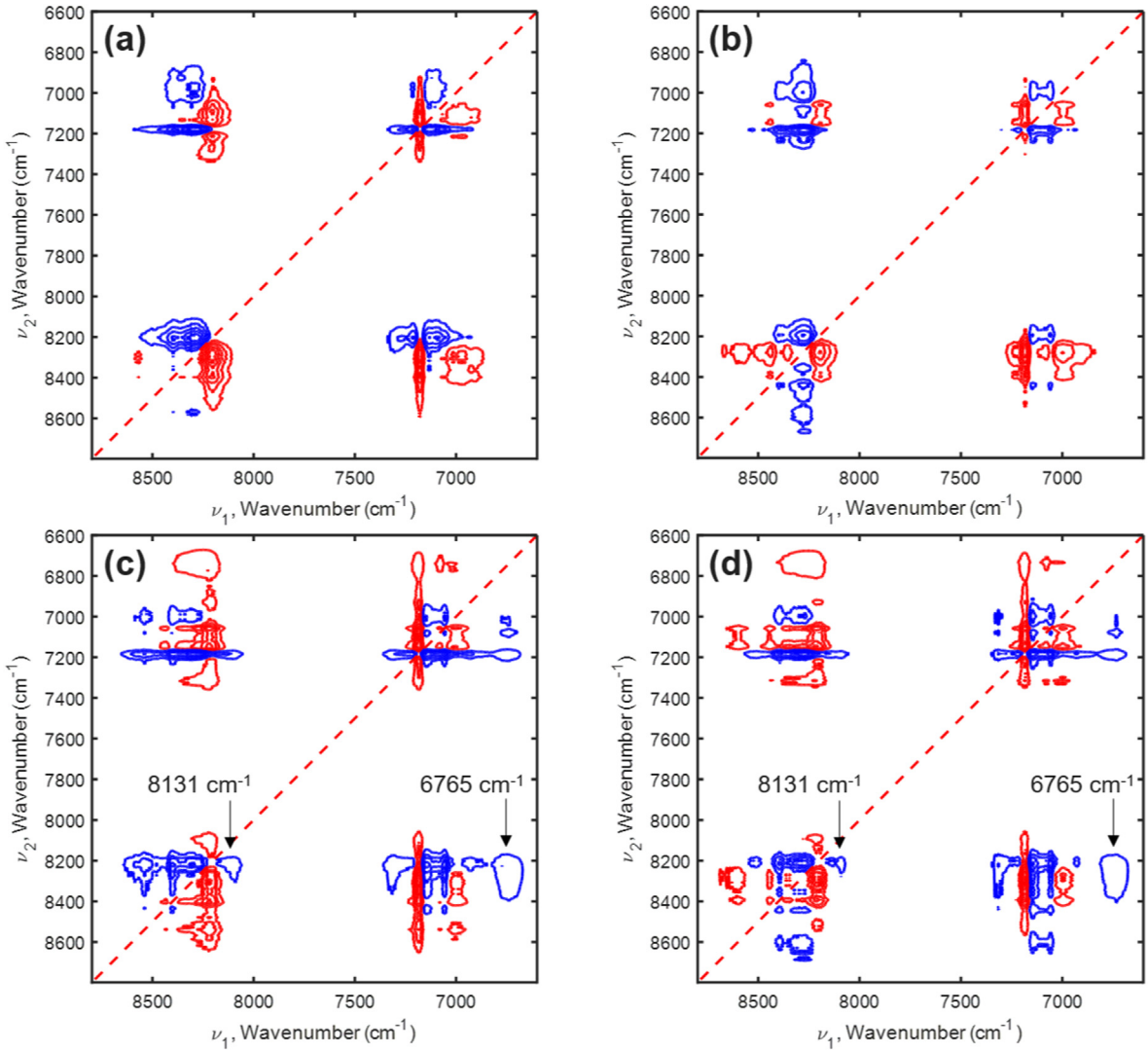

Fig. 3. Asynchronous 2T2D correlation plots obtained from pure oil sample #2 and three adulterated samples commonly based on olive oil #2 (olive oil #2−canola oil #1 (b), olive oil #2−soybean oil #1 (c), and olive oil #2−corn oil #1 (d)).

compared with using the power spectra.

Although each slice spectrum contains own discriminant feature as shown above, use of the whole 2T2D slice spectra utilizing all diverse and mutually characteristic 2T2D spectral information is potentially advantageous to improve the authentication accuracy. Therefore, the serially-merged whole 2T2D slice spectra as well as raw spectra collected at 20 and 41 °C were used for the authentication. Also, the PC scores obtained from the whole slice spectra and raw spectra were also adopted as the input for the evaluation. Next, *k*-NN, PLS-DA, and SVM were employed for the two-group discrimination with two-level CV. All necessary parameters in each method were determined by maximizing accuracy during an inner CV loop and then subsequent accuracy was obtained from testing data set of outer CV loop using determined parameters. The authentication results including accuracy, sensitivity, and specificity for all the cases are summarized in Table 1. TP, FP, TN, and FN denote true positive, false positive, true negative, and false negative samples in the authentication. The selected parameters used for each discrimination method are summarized in the supplementary information.

Initially, when the uses of raw spectra collected at 20 and 41 °C are compared, the accuracies in the 41 °C measurement are

Table 1
Accuracy, sensitivity, and specificity resulted in the authentication of pure olive oils when using the raw spectra, PC scores from the raw spectra, and whole 2T2D slice spectra for *k*-NN, PLS-DA, and SVM.

Input	Method	Accuracy[a]	Sensitivity[b]	Specificity[c]
Raw spectra (20 °C)	*k*-NN	91.1%	22.9%	97.9%
	PLS-DA	94.8%	42.3%	100.0%
	SVM	97.5%	75.6%	99.7%
PCA scores (20 °C)	*k*-NN	91.4%	21.8%	98.4%
	SVM	96.2%	70.1%	98.8%
Raw spectra (41 °C)	*k*-NN	90.5%	14.8%	98.1%
	PLS-DA	91.7%	14.7%	99.4%
	SVM	92.9%	39.2%	98.2%
PCA scores (41 °C)	*k*-NN	90.4%	13.7%	98.1%
	SVM	91.4%	18.9%	98.6%
Whole 2T2D slice spectra	*k*-NN	100.0%	100.0%	100.0%
	PLS-DA	97.8%	81.2%	99.5%
	SVM	98.3%	86.2%	99.5%

[a] Calculated using (TP + TN)/(TP + FN + FP + TN) × 100.
[b] Calculated using (TP)/(TP + FN) × 100.
[c] Calculated using (TN)/(FP + TN) × 100.

generally inferior to those in the 20 °C measurement. The sensitivities of the 41 °C measurement are particularly poor such as all below 39.2%, indicating the inaccurate identification of pure olive oil. Meanwhile, when the raw spectra collected at 20 °C are used for SVM, the accuracy improves to 97.5% with the sensitivity of 75.6%. The substantial improvement of sensitivity is mostly responsible for the enhanced accuracy. This observation informs that the sample spectra collected at 20 °C contains the more discriminant features for the authentication. Alternatively, when the whole 2T2D slice spectra are inputted for *k*-NN, the accuracy reaches to 100%. Simultaneously, the accuracies as well as sensitivities obtained using PLS-DA and SVM are comparatively superior. This comparison informs that the temperature perturbation inducing dissimilar structural variations of components in the pure and adulterated olive oils and the 2T2D slice spectra highlighting the subsequent spectral features are obviously advantage to improve the authentication accuracy.

3.2. NIR spectroscopic discrimination of geographical origins of MVR samples using 2T2D correlation analysis

Fig. 4 (a) presents the average diffuse reflectance NIR spectra of imported (black) and domestic MVR (red) samples in the 9000–4000 cm^{-1} range. The same-colored shade in each average spectrum indicates the standard deviation, noting the magnitude of the sample-to-sample spectral variation in each group. The overall shapes of both the average spectra are similar, because the chemical compositions of the samples from both the geographical origins are not significantly different as expected; however, minute spectral differences exist. Thus, effective recognition of small dissimilar spectral features depending on the geographical origins is critical to achieving accurate discriminant analysis. Fig. 4 (b) shows the scatter plot constructed using the first and second PC scores obtained from all the MVR sample spectra. The black and red circles denote the imported and domestic samples, respectively. Both the groups are identifiable in the score domain, although their scores partially overlap in the middle.

Similar to the case of olive oil authentication, the 2T2D slice spectra at 7925 cm^{-1} of the samples (Fig. 4 (c)) were also chosen as an example and the corresponding 2D score scatter plot (Fig. 4 (d)) was analyzed. The average spectrum of the domestic samples was used as the reference spectrum for the 2T2D correlation analysis. It can be seen that the asynchronous correlation features in the slice spectra of the imported samples (black spectra) are more distinct, such as the sharp peaks at 8176 and 6040 cm^{-1}, and their intensities are higher than those in the domestic samples (red spectra). This is owing to the use of the average spectrum of the domestic samples for comparison. It is reasonable to infer that the 2T2D correlation analysis is effective for highlighting the dissimilar spectral features in the two compared spectra. Therefore, both the groups are more separable and apparent in the subsequent 2D score domain (Fig. 4 (d)), compared to the use of raw spectra.

Table 2 provides the accuracies, sensitivities, and specificities resulted in the discrimination of geographical origin of MVR

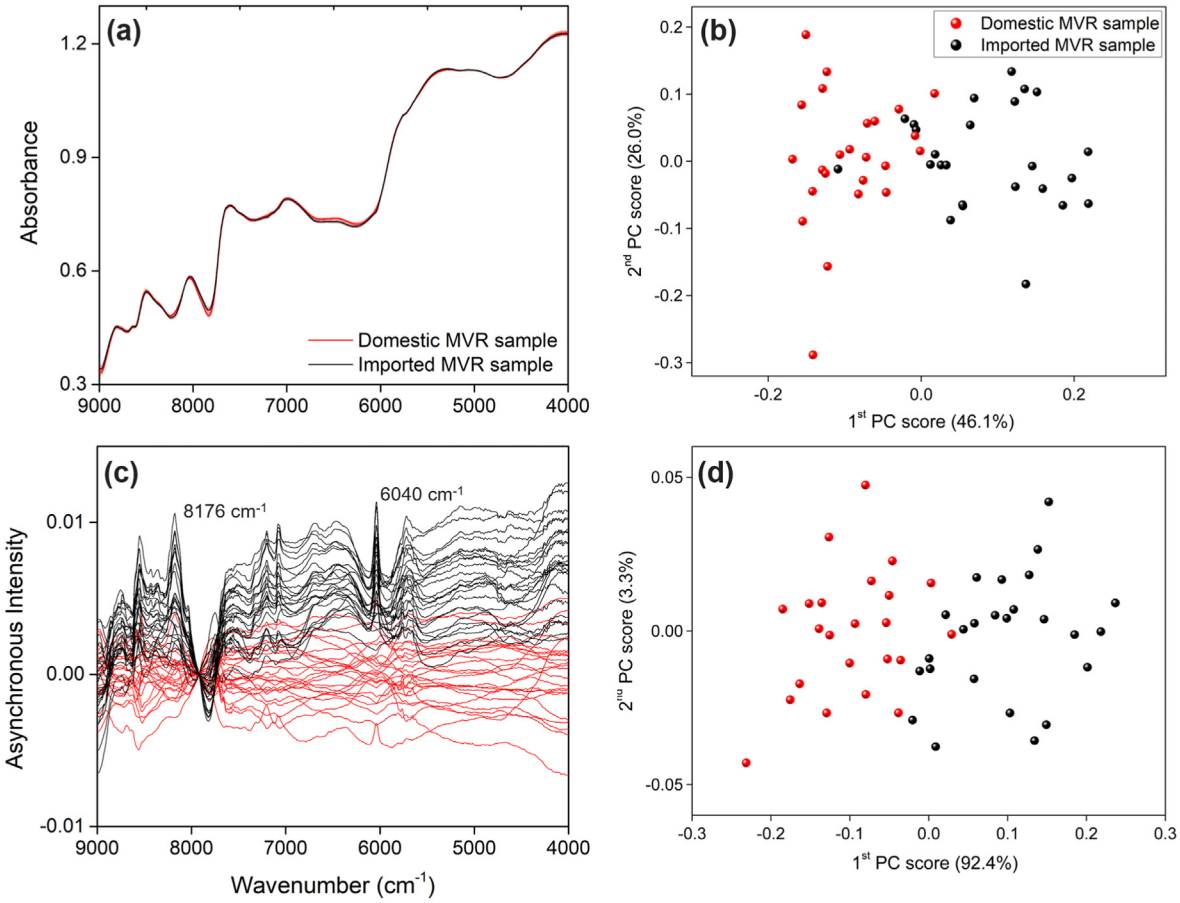

Fig. 4. Average NIR spectra of the imported (black) and domestic MVR (red) samples (a) and the subsequent scatter plot constructed using the first and second scores (b). The same-color shade along with each average spectrum denotes the standard deviation (sample-to-sample spectral variation). Furthermore, the 2T2D slice spectra of all the MVR samples (c) and the subsequent scatter plot (d) are shown. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Accuracy, sensitivity, and specificity resulted in the identifications of geographical origins of MVR and perilla seed samples when using the raw spectra, PC scores from the raw spectra, and whole 2T2D slice spectra for *k*-NN, PLS-DA, and SVM.

Input	Method	Accuracy	Sensitivity	Specificity
MVR				
Raw spectra	*k*-NN	89.1%	90.7%	87.4%
	PLS-DA	93.9%	92.2%	95.6%
	SVM	92.9%	94.2%	91.6%
PCA scores	*k*-NN	90.1%	89.8%	90.4%
	SVM	92.1%	91.2%	92.9%
Whole 2T2D slice spectra	*k*-NN	92.8%	94.6%	91.0%
	PLS-DA	94.8%	95.7%	93.9%
	SVM	93.8%	93.1%	94.5%
Perilla seed				
Raw spectra	*k*-NN	92.9%	94.7%	91.7%
	PLS-DA	93.9%	98.0%	91.5%
	SVM	93.8%	94.2%	93.6%
PCA scores	*k*-NN	92.7%	94.5%	91.7%
	SVM	93.7%	94.8%	93.1%
Whole 2T2D slice spectra	*k*-NN	93.7%	95.5%	92.6%
	PLS-DA	94.4%	98.1%	92.2%
	SVM	93.7%	96.2%	92.2%

samples. Also, refer to the supplementary information for the selected parameters in each discrimination method. The accuracies are better in the cases of using either the raw or whole 2T2D slice spectra for PLS-DA, compared to *k*-NN or SVM. The most accurate discrimination (accuracy: 94.8%) is realized when the whole 2T2D slice spectra are fed for PLS-DA. Based on the observation, the 2T2D slice spectral features obtained by using the average spectrum of the domestic samples as the reference spectrum are more descriptive for the discrimination and PLS-DA is more effective to recognize them, thereby improving the accuracy.

3.3. NIR spectroscopic discrimination of geographical origins of perilla seed samples using 2T2D correlation analysis

The total number of MVR samples used in the above study was only 50; therefore, another study employing a sample set containing numerous and diverse samples is necessary and will be more supportive to demonstrate the potential of 2T2D correlation analysis. Therefore, a sample set incorporating 154 domestic and 91 imported perilla seed samples with wide compositional variation was adopted. Fig. 5 (a) shows the average NIR spectra of the imported (black) and domestic perilla (red) samples in the 9000−4000 cm^{-1} range. As before, the spectral features of both the average spectra are similar, and only the small differences are observed. The 2D score scatter plot obtained from all the sample

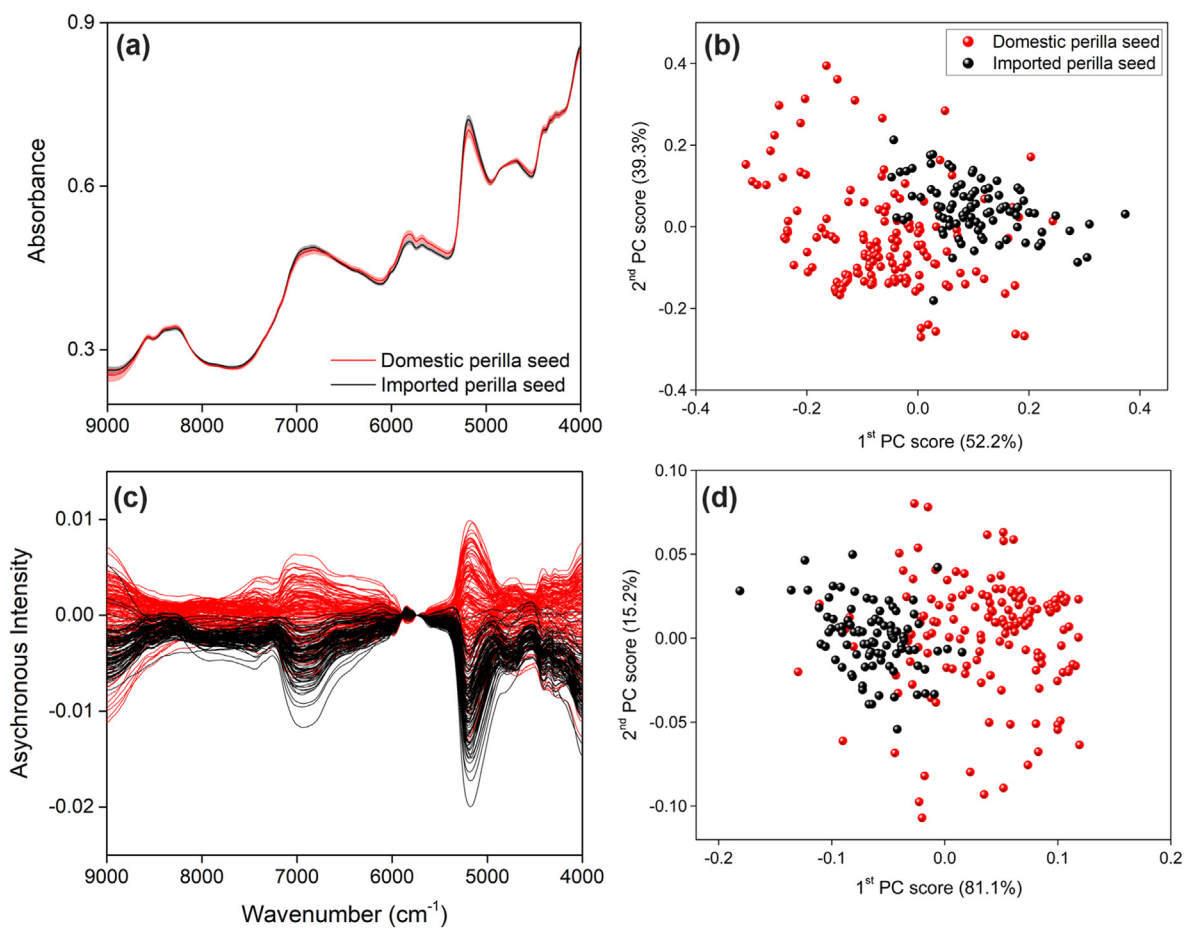

Fig. 5. Average NIR spectra of the imported (black) and domestic perilla seed (red) samples (a) and the subsequent scatter plot constructed using the first and second scores (b). The same-color shade along with each average spectrum denotes the standard deviation. Furthermore, the 2T2D slice spectra of all the perilla seed samples (c) and the subsequent scatter plot (d) are shown. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

spectra is illustrated in Fig. 5 (b). The scores of imported and domestic samples are generally located at the right and left, respectively; however, they overlap considerably at the middle.

Fig. 5 (c) and (d) present the 2T2D slice spectra at 5740 cm^{-1} of all the perilla seed samples (arbitrarily selected) and the subsequent 2D score scatter plot, respectively. As before, the average spectrum of the domestic samples was used as the reference spectrum for the 2T2D correlation analysis. As can be observed, the asynchronous correlation intensities in the imported sample spectra (black) are mostly negative; however, they significantly vary from positive to negative in the domestic sample spectra (red). This suggests that the compositions of the imported samples are relatively similar and their variations are smaller than those of the domestic samples. Although complete distinction of both the groups is not achievable, the two-group presence is more noticeable in the 2D score domain, compared to the case using the raw spectra.

Table 2 also shows the accuracies, sensitivities, and specificities resulted in the discrimination of geographical origin of perilla seed samples. The selected parameters in each case are shown in the supplementary information. The similar results are also observed in the discrimination of perilla seed samples. The accuracies obtained using PLS-DA are generally better than those employing *k*-NN or SVM, and the discrimination becomes most accurate (accuracy: 94.4%) with utilizing the whole 2T2D slice spectra. Again, the 2T2D slice spectral features provide more discriminant information according to the samples' geographical origins and PLS-DA is better matched to improve the accuracy by featuring out the relevant information.

4. Conclusion

In this study, 2T2D correlation analysis effectively highlighted the dissimilar temperature-driven spectral characteristics of pure and adulterated olive oil samples in the subsequent slice spectra, and thereby assisted in improving the accuracy of olive oil authentication. In addition, another strategy of employing the average spectrum of one sample group as the reference spectrum in 2T2D correlation analysis for a two-group discrimination was simple and feasible to improve the accuracy of identifying the geographical origins of the MVR and perilla seed samples. This result demonstrates the potential analytical utility of 2T2D correlation analysis for its application in general discriminant analysis, because it does not require time-consuming and difficult external perturbations, such as temperature variation. For further validation of 2T2D correlation analysis as a potential feature-descriptor, diverse case studies are needed. Therefore, our on-going investigations include a study using a dataset with different perturbations, such as voltage-applied Raman spectra of biological samples, and another study on hetero-2T2D correlation analysis using two different spectroscopic data, such as the IR and Raman spectra of a sample.

CRediT authorship contribution statement

Woosuk Sohng: Data curation, Validation, Visualization. **Changhwan Eum:** Investigation, Formal analysis. **Hoeil Chung:** Conceptualization, Writing - original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aca.2021.338255.

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